Question 1 : Explain the fundamental differences between DDL, DML, and DQL commands in SQL. Provide one example for each type of command.

- DDL- Data Definition language. It changes the structure of the Data. Used to create, delete or modify database objects such as- Tables, Databases, views, indexes.

  Command: Create, Drop, alter, Truncate etc.

- DML- Data Manipulation language. It is used to manipulate the data inside the tables Commands- Insert, delete, update
- DQL- Data Query language. It is used to select the Data from the Database. Command: Select

Question 2 : What is the purpose of SQL constraints? Name and describe three common types of constraints, providing a simple scenario where each would be useful.

Ans: SQL constarints are used to maintain data integrity and consistency and it helps to prevent the invalid data being stored.

Three common type of consrtaints are-

Primary Key Constraints:

It uniquely identifies each row in a table.

Rules-

- Must be unique
- Can not contain null values
- One primary key in the table

```
CREATE TABLE Customers (
   Customer_ID INT PRIMARY KEY,
   Customer_Name VARCHAR(50)
);
```

So here primary key make sure that no two customer got the same customer id.

Foreign key Constarints:

A foreign key creates relationship between two tables.

A value in one table must exist in other table.

```
Create Table customer (

  Customer_id  PRIMARY KEY,

   Customer_name  VARCHAR(50)

);
```

```
create   table orders(
order_id int primary key,
customer_id int ,
foreign key(customer_id)
references customer(customer_id));
```

if the customer id is not in the customer table then that invalid customer_id will not be added to the order tables.

Question 3 : Explain the difference between LIMIT and OFFSET clauses in SQL. How would you use them together to retrieve the third page of results, assuming each page has 10 records?

Ans: Limit – how many rows you want to retrieve from the result.

Offset- How many rows to skip

```
# to select the only first 10 rows
select * from
customer
limit 10;

## to skip the first 10 rows
select   * from customer
offset 10;

## Now to get the 3rd page data when each page contains only  10 rows

select * from
customer
limit 30
offset   20;
```

Question 4 : What is a Common Table Expression (CTE) in SQL, and what are its main benefits? Provide a simple SQL example demonstrating its usage.

Ans: CTE- common table expression is a temporary table created in a single query to break the complex logic into smaller parts. This tables is created using the "WITH" keyword.

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```
with spending as

(select customer_id, round(sum(amount),2) as Total_amount_spent

from Payment

group by Customer_id )


select  C.customer_id,  C.first_name,C.last_name, C.email, S.Total_amount_spent

from customer C Left join spending S

on C.customer_id= S.customer_id

order by Total_amount_spent desc

limit 5;
```

here you can see that a CTE "Spending " is created first to where Total amount spend has been calculated after that in that same query using that 'spending' table we can get top 5 customer who spend the most.